

Exercises: Matrices

$$A = \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 1 \\ -1 & 5 \\ 4 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 5 & 2 \\ -1 & 4 & 0 \end{pmatrix}$$

$$D = \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix}$$

Question 1a

$$A + A$$

Answer:

$$\begin{aligned} A + A &= \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} + \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 8 \\ -4 & -10 \end{pmatrix} \end{aligned}$$

Question 1b

$$3D$$

Answer:

$$\begin{aligned} 3D &= 3 \cdot \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 12 & 9 & 6 \\ -3 & 15 & 0 \\ 6 & 12 & 0 \end{pmatrix} \end{aligned}$$

Question 2a

$$AA$$

Answer:

$$\begin{aligned} A \cdot A &= \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \cdot \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \\ &= \begin{pmatrix} (2 \cdot 2 + 4 \cdot (-2)) & (2 \cdot 4 + 4 \cdot (-5)) \\ (-2 \cdot 2 + (-5) \cdot (-2)) & (-2 \cdot 4 + (-5) \cdot (-5)) \end{pmatrix} \\ &= \begin{pmatrix} -4 & -12 \\ 6 & 17 \end{pmatrix} \end{aligned}$$

Question 2b

$$AC$$

Answer:

$$A \cdot C = \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 5 & 2 \\ -1 & 4 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & 26 & 4 \\ 3 & -30 & -4 \end{pmatrix}$$

Question 2c

BA

Answer:

$$\begin{aligned} B \cdot A &= \begin{pmatrix} 2 & 1 \\ -1 & 5 \\ 4 & 6 \end{pmatrix} \cdot \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 3 \\ -12 & -29 \\ -4 & -14 \end{pmatrix} \end{aligned}$$

Question 2d

BC

Answer:

$$\begin{aligned} B \cdot C &= \begin{pmatrix} 2 & 1 \\ -1 & 5 \\ 4 & 6 \end{pmatrix} \cdot \begin{pmatrix} 1 & 5 & 2 \\ -1 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 14 & 4 \\ -6 & 15 & -2 \\ -2 & 44 & 8 \end{pmatrix} \end{aligned}$$

Question 2e

CD

Answer:

$$\begin{aligned} C \cdot D &= \begin{pmatrix} 1 & 5 & 2 \\ -1 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 3 & 36 & 2 \\ -8 & 17 & -2 \end{pmatrix} \end{aligned}$$

Question 2f

DB

Answer:

$$\begin{aligned} D \cdot B &= \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ -1 & 5 \\ 4 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 13 & 31 \\ -7 & 24 \\ 0 & 22 \end{pmatrix} \end{aligned}$$

Question 2g

DD

Answer:

$$\begin{aligned} D \cdot D &= \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 3 & 2 \\ -1 & 5 & 0 \\ 2 & 4 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 35 & 8 \\ -9 & 22 & -2 \\ 4 & 26 & 4 \end{pmatrix} \end{aligned}$$

Question 3

There is one last possible matrix multiplication involving two of these matrices. What is it? What is the result of the multiplication?

Answer:

$C \cdot B$

$$\begin{aligned} C \cdot B &= \begin{pmatrix} 1 & 5 & 2 \\ -1 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ -1 & 5 \\ 4 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 38 \\ -6 & 19 \end{pmatrix} \end{aligned}$$

Question 4

Calculate $\det(A)$, and using it, calculate A^{-1}

Answer:

$$\text{If } A = \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix}$$

$$\begin{aligned} \det(A) &= (2)(-5) - (4)(-2) \\ &= -10 - (-8) \\ &= -2 \end{aligned}$$

$$\begin{aligned} A^{-1} &= -\frac{1}{2} \cdot \begin{pmatrix} -5 & -4 \\ 2 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 5/2 & 2 \\ -1 & -1 \end{pmatrix} \end{aligned}$$

Question 5

Using the result of the previous question, solve the following system of linear equations:

$$\begin{aligned} 2x_1 + 4x_2 &= 22 \\ -2x_1 - 5x_2 &= -26 \end{aligned}$$

Answer:

A system of linear equations can be expressed as $A\vec{x} = \vec{b}$ and the solution to this can be expressed as $\vec{x} = A^{-1}\vec{b}$

Writing these equations as matrices:

$$\begin{aligned} A\vec{x} &= \vec{b} \\ \begin{pmatrix} 2 & 4 \\ -2 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} 22 \\ -26 \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
\vec{x} &= A^{-1}\vec{b} \\
&= \begin{pmatrix} 5/2 & 2 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 22 \\ -26 \end{pmatrix} \\
&= \begin{pmatrix} 3 \\ 4 \end{pmatrix}
\end{aligned}$$

Therefore $x_1 = 3$ and $x_2 = 4$.